

Jaisidh Singh

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🌐 Website 🐙 Github in LinkedIn 📄 Google Scholar

Education

- 2024–2026 **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
GPA: 1.52/5.0 (German scale, 1.0 to 5.0 from best to worst)
Thesis (in association with OpenEuroLLM): architecture scaling laws for hybrid attentions LLMs.
- 2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.
Thesis: negation-aware compositional reasoning in vision-language models.

Awards & fellowships: Konrad Zuse School (ELIZA) Fellowship for Master's Students 2024-2026.

Selected Publications

- 2025 **(Almost) Free Modality Stitching of Foundation Models**, *EMNLP'25, ICLR-W'25*, Jaisidh Singh, Diganta Misra, Boris Knyazev, Antonio Orvieto, **(66 2 citations)**.
- 2025 **Learning the Power of “No”: Foundation Models with Negations**, *WACV'25*, Jaisidh Singh*, Ishaan Shrivastava*, Mayank Vatsa, Richa Singh, Aparna Bharati, **(66 50 citations)**.
- (Refer to Google Scholar for a complete list of publications, * indicates equal contribution.)

Work/Research Experience

- Oct 2025 – **Software Engineer Graduate Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
May 2026
 - Created real-time interactive computer vision tools for Stadtmuseum Tübingen. Designed visitor UX flow to flexibly meet domain constraints, informing latency-quality tradeoff optimization.
 - Developed & delivered a tutorial on LLM-agent orchestration with LangGraph at GCPR 2025.
- Jan 2025 – **Guest Researcher at MPI-IS Tübingen**, *With Dr. Antonio Orvieto*.
present
 - Analyzed transformer generalization dynamics & accelerated grokking speed by upto 8×.
 - Created a hypernetwork for efficient (10×) modality stitching for CLIP and multimodal LLMs. Published at EMNLP 2025.
- Winter 2022 & 2023 **Researcher at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa & Prof. Richa Singh*.
 - Researched negation understanding in vision-language models. Published at WACV'25.
 - Analysed identity leakage in GAN latent space. Published at WACV'24.
- Summer 2022 & 2023 **Research Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.
 - Devised an interpretable failure discovery system, accurate upto 95%, for vision models.
 - Developed image retrieval pipelines for internal experiments.

Selected Projects & Releases

- loracclip**: python library to cleanly wrap adding LoRA to CLIP. **(40 ☆ on 🐙)**
- pytorch-mixtures**: a minimalist library for MoEs in PyTorch. **(28 ☆ on 🐙)**

Technical Skills

- Languages**: English (Fluent + Professional), Hindi (Native), German (A1)
- Machine learning**: JAX, FLAX, Optax, PyTorch, Huggingface
- Distributed training & HPC tools**: Slurm, Condor, Data/Model/Tensor Parallel
- Programming languages**: Python, $\LaTeX$, Bash, JavaScript, C++

Academic Service

- Reviewer**: CVPR'26, ECCV'26, NeurIPs'26.
- Teaching assistant**: Course: Understanding Language Models (Summer Semester '26 at University of Tübingen, with Prof. Carsten Eickhoff).